

# RESEARCH SUMMARY

Songyao Xue | Fall 2027

*How can imperfect quantum resources be transformed into reliable computational advantages?*

**Research focus.** My work examines how noise, limited entanglement, protocol exposure, and finite computational budgets can be modeled and reorganized into secure, resource-efficient, and adaptive quantum information systems. My background in Software Engineering provides a computational perspective grounded in algorithms, scientific programming, reproducibility, and implementation cost.

## RESEARCH TRAJECTORY

### 1. Structural quantum security

Analyzed how Grover-based verification structure can enable key recovery, forgery, and repudiation. Contributed to four reusable protocol-inspection points and a strengthened arbitrated signature design using teleportation and strengthened QOTP.

### 2. Resource-efficient probabilistic decoding

Reinterpreted Huffman lengths as support budgets in a common Hilbert space; combined group ancilla, conditional POVMs, QEC reliability, MAP decisions, and SDP-based resource certificates. The central result is that improved reliability is meaningful only when it survives explicit overhead accounting.

### 3. Weak-link quantum communication

Developed a no-signaling framework in which weak, heterogeneous entanglement links are treated as task-aware resources. Derived AND/OR first-order corrections, noise-to-visibility mappings, Bayesian weighting, and cross-platform calibration rules for photonic, trapped-ion, and superconducting settings.

### 4. Learnable quantum networks

Proposed a graph-based weak-link multi-group quantum game framework with game tensors, sign rules, spectral bounds, and differentiable relaxations. Implemented automated search, repeated trials, topology/noise comparisons, and backend-side circuit checks while separating theorem-level results from surrogate and proxy quantities.

## SELECTED METHODS AND PREPARATION

- Quantum protocol and information-flow analysis; graph-based weak-link modeling; first-order and spectral analysis.
- POVM-based state discrimination, semidefinite programming, resource-normalized criteria, Bayesian and MAP decisions.
- Python scientific computing, Qiskit, PennyLane, automated experiments, sensitivity analysis, and visualization.
- LaTeX manuscript preparation, appendix proofs, figure/table generation, reference management, and reproducible research organization.
- B.Eng. in Software Engineering, academic rank Top 10%; supervisor-led reading group in quantum information, quantum communication, cryptography, and quantum computing.

## CURRENT RESEARCH OUTPUTS

- On Grover-Based Quantum Signature Schemes — second author; revised and resubmitted.
- A Learnable Unified Framework for Weak-Link Multi-Group Quantum Games — first author; under editorial review.
- Weak-Link Entanglement-Assisted No-Signaling Quantum Communication Protocols — first author; manuscript in preparation.
- Error-Corrected Quantum-Huffman-Inspired Probabilistic Source Decoding — first author; manuscript in preparation.

## FUTURE RESEARCH AGENDA

**Toward a resource-aware Quantum Internet.** I aim to study quantum-network protocols that jointly optimize topology, routing, entanglement quality, security, and task performance; learnable graph-structured quantum systems that adapt under calibrated noise; and unified distributed architectures in which communication, computation, cryptography, and error correction are evaluated under the same finite-resource budget.

Quantum Information · Quantum Communication · Quantum Cryptography · Quantum Networks · Quantum Machine Intelligence

Luoyang Normal University | 18637229290@139.com